The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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Choosing Your Software

Software	Category	License	Why it's interesting
VLC Media Player	Multimedia	`LGPL / GPL`	*Plays anything — built by students in Paris*

PROJECT REPORT

PART A — ORIGIN & PHILOSOPHY (VLC)

VLC Media Player was created in the late 1990s by a group of students in Paris under the VideoLAN project. The main aim at that time was to build something that could stream video over a university network, because there were no good free tools available for that purpose. Most of the software that existed either didn't support streaming properly or was not accessible to students since it was paid or restricted.

Before VLC came into the picture, people faced a lot of issues while playing media files. One major problem was compatibility. Different media players supported different formats, so users often had to install multiple applications just to play different types of videos or audio files. This made things confusing and inconvenient. Also, streaming video over a network was not simple at all and usually required complex setups or expensive software. Another issue was that users had almost no control over codecs or playback features, because most software was closed source and could not be modified.

Because of these problems, the students decided to create their own solution that was flexible and could handle multiple formats easily. They also wanted something that could be improved over time, so they made VLC open source. This allowed other people to study the code, modify it, and contribute to it. As more developers got involved, VLC kept improving and became more powerful. Over time, it turned into a widely used media player that can play almost any format and is trusted by users all over the world.

VLC wasn't just a player — it solved the problem of media fragmentation, it gave users control over what and how they play content.

1. Four freedoms

- Freedom to run program
- Freedom to study code
- Freedom to modify
- Freedom to distribute

VLC:

Yes, it supports all four (because of GPL/LGPL)

2. GPL vs MIT

- Choose GPL if you want freedom preserved
- Choose MIT if you want maximum adoption

3. Free beer vs free speech

- Free beer = free of cost
- Free speech = freedom to modify

VLC example:

- VLC is both:
    - Free to download
    - And you can modify its code

Open source **accelerates innovation**, not kills it

PART B — LINUX FOOTPRINT (VLC)

Installation

On Ubuntu:

```
sudo apt install vlc
```

Key directories

- Binary:
  /usr/bin/vlc
- Config:
  ~/.config/vlc
- Libraries:
  /usr/lib/vlc

Updates

- Via package manager:

```
sudo apt update
sudo apt upgrade
```

PART C — ECOSYSTEM

Dependencies

- FFmpeg (codecs)
- libdvdcss
- Qt (UI framework)

What VLC enabled

- Universal media playback
- Streaming tools
- Media conversion

Relation to web

- Not part of LAMP directly
- BUT used in:
    - Streaming
    - Media servers

Community

- Developed by **VideoLAN organization**
- Contributors worldwide
- Forums, GitLab, mailing lists

PART D — COMPARISON

VLC vs Windows Media Player

Dimension	VLC	Windows Media Player
Cost	Free	Comes with Windows
Security	Open audit	Closed source
Support	Community	Microsoft support
Flexibility	High	Limited
Control	User-driven	Corporate-controlled

Final Verdict

- VLC is powerful, flexible, and open
- Best for users who want control

Script 1 — System Identity Report

Terminal

```
$ #!/bin/bash
# Script 1: System Identity Report
# Author: Your Name | Course: Open Source Software

# --- Variables ---
STUDENT_NAME="Your Name"
SOFTWARE_CHOICE="VLC Media Player"

# --- System info ---
KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)
HOME_DIR=$HOME
DISTRO=$(grep '^PRETTY_NAME' /etc/os-release | cut -d= -f2 | tr -d '"')

# --- Display ---
echo "========================================"
echo "  Open Source Audit — $STUDENT_NAME"
echo "========================================"
echo "Software Chosen : $SOFTWARE_CHOICE"
echo "Distro          : $DISTRO"
echo "Kernel          : $KERNEL"
echo "User            : $USER_NAME"
echo "Home Directory  : $HOME_DIR"
echo "Uptime          : $UPTIME"
echo "Date & Time     : $DATE"
echo ""
echo "License Info: This system is based on open-source software (GPL licensed Linux kernel)."
echo "========================================"
```

Script 2 — FOSS Package Inspector

Terminal

```
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="vlc"

# Check if package is installed (Debian/Ubuntu)
if dpkg -l | grep -qw $PACKAGE; then
    echo "$PACKAGE is installed."
    dpkg -l | grep $PACKAGE
else
    echo "$PACKAGE is NOT installed."
fi

# Case statement for philosophy note
case $PACKAGE in
    vlc) echo "VLC: a free media player that supports all formats and promotes openness" ;;
    firefox) echo "Firefox: a browser built to protect user privacy and open web standards" ;;
    mysql) echo "MySQL: open-source database powering modern applications" ;;
    git) echo "Git: distributed version control enabling collaborative development" ;;
    *) echo "Unknown package" ;;
esac
```

Script 3 — Disk and Permission Auditor

```
Terminal

#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist on this system"
    fi
done

echo ""
echo "Checking VLC config directory..."

VLC_CONFIG="$HOME/.config/vlc"

if [ -d "$VLC_CONFIG" ]; then
    ls -ld $VLC_CONFIG
else
    echo "VLC config directory not found"
fi
```

Script 4 — Log File Analyzer

```
Terminal

#!/bin/bash
# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh /var/log/syslog error

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "Error: File $LOGFILE not found."
    exit 1
fi

# Retry if file is empty (do-while style)
while [ ! -s "$LOGFILE" ]; do
    echo "File is empty. Waiting for content..."
    sleep 2
done

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

echo ""
echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5
```

Script 5 — Manifesto Generator

```
>_ Terminal

#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer three questions to generate your manifesto."
echo ""

read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# Creating manifesto paragraph
echo "On $DATE, I reflect on the importance of open source." > $OUTPUT
echo "Using tools like $TOOL, I understand that freedom means $FREEDOM." >> $OUTPUT
echo "In the future, I aspire to build $BUILD and share it with the world." >> $OUTPUT
echo "Open source allows innovation, collaboration, and transparency." >> $OUTPUT

# Alias concept (example in comment)
# alias manifesto='cat $OUTPUT'

echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```

This script uses variables, command substitution, and echo statements to display system information such as kernel version, user, and uptime. It demonstrates basic shell scripting concepts like variable assignment and formatted output.

Project Timeline

Day	Focus	Tasks	Deliverable
1	Setup	Selected VLC Media Player. Researched origin and license. Set up Linux environment. Installed VLC using package manager.	Choice confirmed
2	Philosophy	Studied the origin of VLC and the problem it solved. Wrote Part A1 in own words.	Draft A1
3	License & Ethics	Analyzed GPL/LGPL license. Studied four freedoms. Reflected on ethical questions.	Draft A2 + A3
4	Linux Footprint	Explored installation directories, permissions, and update process. Took screenshots.	Draft Part B
5	Scripts 1 & 2	Developed System Identity script and Package Inspector script. Tested both on Linux.	Scripts 1–2 working
6	Scripts 3 & 4	Created Disk Auditor and Log Analyzer scripts. Tested with real system logs.	Scripts 3–4 working
7	Ecosystem	Researched VLC dependencies, tools, and community ecosystem.	Draft Part C
8	Comparison & Script 5	Compared VLC with Windows Media Player. Created table and wrote Manifesto script.	Draft Part D + Script 5
9	Report polish	Compiled full report. Added introduction, conclusion, and screenshots. Checked page count.	Complete draft
10	Submission	Reviewed all work. Uploaded scripts to GitHub. Submitted report on portal.	Final submission

7. Reference Resources

On Open Source Philosophy

- GNU Project — *The Free Software Definition*
- Open Source Initiative — *The Open Source Definition*
- Eric S. Raymond — *The Cathedral and the Bazaar*
- Linus Torvalds — *Just for Fun* (Autobiography)

On Linux and Shell Scripting

- *The Linux Command Line* by William Shotts
- GNU Bash Manual
- Linux man pages (man <command> in terminal)

License Information

Official repository of VLC Media Player

https://code.videolan.org/videolan/vlc

These resources were used to study the licensing model of VLC Media Player, including the GPL and LGPL licenses. The official repository provided access to the LICENSE file.

Conclusion

In this project, I explored the concept of open-source software through the example of VLC Media Player. I learned how VLC was created to solve real-world problems related to media compatibility and streaming, and how its open-source nature allowed it to grow into a widely used tool. Through studying its license, I understood the importance of software freedom and the difference between free access and user control.

I also gained practical experience by working on Linux and writing shell scripts, which helped me understand how software interacts with the operating system. The comparison between open-source and proprietary software showed that while open source offers flexibility and transparency, both models have their own advantages depending on the use case. Overall, this project helped me appreciate the value of open-source software and how it contributes to innovation and collaboration in the tech industry.